

Computational fluid dynamics coupled to biokinetic models: Numerical methodology for microalgae cultivation optimization

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ABSTRACT

Mixing of the algal culture medium in a photobioreactor (PBR) is a crucial factor to enhance microalgae cultivation and production, since it ensures microalgae cells can homogeneously access nutrients and light radiation. Hydrodynamic conditions in an industrially applicable PBR cannot be tested in laboratory cultivation systems since the scale-up methodology is not specified; hence requiring complex tests at a demonstrative scale or the application of mathematical models. In this study, a three-dimensional numerical model of the hydrodynamic conditions in a hybrid horizontal tubular PBR was developed to investigate the mixing of the culture medium under various operating conditions, in order to optimize biomass production in this type of system. The PBR consists of two open tanks connected through sixteen polyethylene transparent tubes, at a demonstrative scale (total volume 11.7 m³). Microalgae cells movement inside the tubes was simulated. Cell trajectory prediction allows to simulate the intensity of mixing of the culture medium. Subsequent coupling with the biokinetic BIO_ALGAE model allows to monitor the influence of hydrodynamic conditions on the distribution of light in the culture medium and the yield of microalgae. The simulations were experimentally validated. The more intense mixing of the culture medium caused by increasing flow rate ($Re = 23,700$ – $46,200$) allows more frequent exposure of the microalgae cells to the light source, which according to the results obtained can increase the microalgal yield by 14% in the summer cultivation period and by 151% in the winter period.

1. Introduction

Microalgae produce a wide range of high-value products such as lipids, carotenoids, and polyunsaturated fatty acids (Karimian et al., 2022). The industrial cultivation of microalgae is a relatively cost-demanding process, which still restrains its application in the market (Ruiz et al., 2016; Tredici et al., 2016). Cultivation systems using nutrients from wastewater for microalgae cultivation can be applied in a circular economy approach, reducing the operational cost of the production system (Díez-Montero et al., 2020).

The cultivation of microalgae is performed in photobioreactors (PBR), in which a wide range of performances and biomass productivities can be achieved. The variations in productivity can be attributed to several factors, being the hydrodynamics one of the most important, since mixing and hydrodynamic conditions in culture

medium have a strong influence on nutrients and light radiation distribution, and subsequently on microalgae culture performance (Huang et al., 2014). Due to absorption and scattering, light gradients can form in the culture medium layer (Perner-Nochta and Posten, 2007; Sforza et al., 2014), resulting in the formation of light and dark zones (Zhang et al., 2013). In dark zones, the cells do not absorb sufficient light radiation and the yield of biomass can be subsequently reduced due to low active radiation availability (Khan et al., 2018). The hydrodynamic behaviour of a PBR cannot be easily tested at lab or bench scale, since the upscale to pilot or industrial scale is not evident. Therefore, the optimization of industrial microalgae cultivation and production should be performed through demonstrative scale experiments or through mathematical modelling integrating the biokinetic and hydrodynamic phenomena.

Computational fluid dynamics (CFD) can be used for a detail study of hydrodynamic conditions in different photobioreactors (Bitog et al.,

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Nomenclature

COD	($\text{gO}_2 \text{ m}^{-3}$), Chemical oxygen demand.
d	(m), Depth of culture medium in PBR.
d_{tank}	(m), Average depth of the culture medium in the retention tank.
H	(m), Distance of the particle from the irradiated wall.
h_1	(m), Culture medium level in discharge section of retention tank.
h_2	(m), Culture medium level in suction section of retention tank.
I_{av}	(W m^{-2}), Average light intensity.
I_0	(W m^{-2}), Incident light intensity.
I_p	(W m^{-2}), Local irradiance at a given depth.
K_{a,CO_2}	(s^{-1}), Mass transfer coefficient for carbon dioxide.
K_{a,NH_3}	(s^{-1}), Mass transfer coefficient for ammonia.
K_{a,O_2}	(s^{-1}), Mass transfer coefficient for oxygen.
$k_{\text{death},H}$	(s^{-1}), Decay of heterotrophic bacteria.
K_I	($\text{m}^2 \text{ g}^{-1}$), Extinction coefficient for microalgae biomass.
L	(m), Length of the transparent tube.
Re	(-), Reynolds number.

S_{NH_4}	($\text{gNH}_4^+-\text{N m}^{-3}$), Ammonium nitrogen.
S_{O_2}	($\text{gO}_2 \text{ m}^{-3}$), Dissolved oxygen.
t_i	(s), time.
TSS	(gCOD m^{-3}), Total suspended solids.
$\bar{u}$	(m s^{-1}), Mean velocity.
X_{ALG}	(g m^{-3}), Microalgae biomass concentration in culture medium.
X_C	(g m^{-3}), Sum of particulate components (microalgae and bacteria).
Δh	(m), Difference of culture medium level.
μ_{ALG}	(s^{-1}), Maximum growth rate of microalgae.
μ_H	(s^{-1}), Maximum growth rate of heterotrophic bacteria.
ν	($\text{m}^2 \text{ s}^{-1}$), Kinematic viscosity.
ASM3	Activated Sludge Model No. 3.
CFD	Computational fluid dynamics.
DO	Dissolved oxygen.
HHT PBR	Hybrid horizontal tubular photobioreactor.
LES	Large eddy simulation.
RANS	Reynolds Averaged Navier-Stokes.
RMSE	Root mean square error.
RWQM1	River Water Quality Model 1.

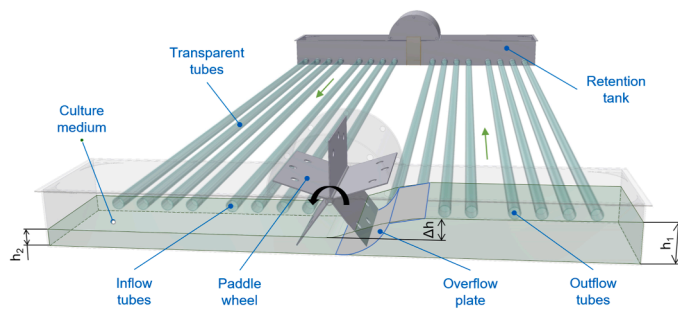


Fig. 1. 3D conical view of HHT PBR.

2011). Loomba et al. (2021) studied the effect of different operational and design arrangements on the movement of microalgae cells in a flat-panel photobioreactor. Based on the developed model, it was possible to simulate the movement of the cells between the illuminated and dark zones of the photobioreactor and the subsequent effect on the production of microalgal biomass.

The results of published studies demonstrate that numerical modelling can also be used to simulate the biochemical processes of microalgae cultivation. Several models have been developed recently at this aim. ALBA model (Casagli et al., 2021) describes growth and interactions among microalgae and bacteria in outdoor raceway systems for wastewater treatment. However, the developed ALBA model assumes complete mixing of the culture medium in the system, which is not the case in full scale cultivation wastewater systems. ABACO model developed by Sánchez-Zurano et al. (2021) was calibrated based on experimental data from a laboratory photobioreactor with 1 L of culture medium, and a complete recalibration of the model is certainly needed when scaling-up. The effect of hydrodynamic conditions on the cultivation process is not included in the model. Solimeno, Parker, et al., (2017) created a mechanistic biokinetic model (named BIO_ALGAE) that describes the complex interactions between microalgae and bacteria in microalgae-based wastewater treatment systems. The model has been successfully used for several applications, such as column photobioreactor, vertical tubular photobioreactor, or hybrid horizontal tubular (Solimeno et al., 2017a). The BIO_ALGAE model considers the decrease of light in the culture medium, and it has been observed light

attenuation can limit microalgae growth by up to 60% (Solimeno et al., 2017b). BIO_ALGAE model works on the assumption that the culture medium is ideally mixed. The decrease in light is thus ideal homogeneous, and the temperature of the culture medium is also homogeneous throughout the entire volume of the PBR. The reactors also work ideally in terms of nutrient utilization and mass transfer.

One of the main drawbacks or limitation of all these above cited models is that the biokinetics assume perfect complete mixing in the reactor, i.e. they have zero-dimensional approaches for biokinetics. Of course, the absence of hydrodynamics influence on microalgal biomass production can be negligible in laboratory equipment. However, the influence of hydrodynamics will be critical when transforming cultivation systems to pilot or industrial scales (Shoener et al., 2019). Fernández del Olmo et al. (2021) integrated the influence of hydrodynamic conditions and variations of photobioreactor tube size into the dynamic photosynthetic model proposed by Rubio Camacho et al. (2003). The model showed that light radiation is limited in systems operating with larger diameter transparent tubes, however, this model was not verified using experimental data from a real equipment.

The aim was to couple the mechanistic biokinetic BIO_ALGAE model with a numerical model simulating the hydrodynamic conditions of the culture medium, in order to achieve more realistic simulations and to optimize the microalgae cultivation process in hybrid horizontal tubular photobioreactor (HHT PBR). For this purpose, a three-dimensional CFD model describing hydrodynamic conditions in HHT PBR was developed. Hydrodynamic model allowed to predict the effect of changing operating conditions on the movement of microalgae cells, while multi-physical modelling integrating the hydrodynamics and biokinetic models allowed predicting the effect of mixing on the intensity of light radiation incident on the microalgae cells and the subsequent effect on the microalgae yield. The applicability of the developed multi-physical model was verified with results from experimental measurements in demonstrative scale HHT PBR.

2. Materials and methods

2.1. Photobioreactor design

The HHT PBR was a demonstrative scale system for the treatment of agricultural and domestic wastewater and the production of microalgal biomass. The HHT PBR was installed in Agròpolis campus of the

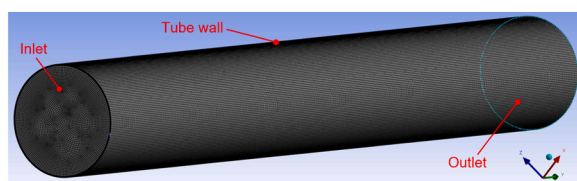


Fig. 2. Boundary conditions defined in the generated mesh of simulated tube geometry.

Table 1

Operational setup in HHT PBR for calibration and validation of the CFD model.

Configuration	h_1 (m)	h_2 (m)	Δh (m)	$\bar{u}$ (m s ⁻¹)	Re (-)
A	0.28	0.24	0.04	0.19	23,700
B	0.35	0.29	0.06	0.25	31,200
C	0.39	0.26	0.13	0.37	46,200

Universitat Politècnica de Catalunya - BarcelonaTech, as part of a more complex installation for resource recovery from wastewater built under European funding (INCOVER project GA 689242). Three HHT PBRs connected in parallel constituted the entire pilot plant (one reactor was used for the purposes of this work). Detailed information about the plant can be found in García et al. (2018).

Each HHT PBR consists of two open tanks that are interconnected by 16 tubes in two rows (Fig. 1). Interconnecting transparent LDPE tubes (0.125 m internal diameter) with a total length of $L = 47$ m define the mutual distance between the open tanks. Paddle wheels are installed in both open tanks to pump the culture medium through an overflow from one side of the tank to the other, thus creating a difference in the level of the culture medium Δh (m), which ensures a pressure differential at both ends of the tubes. The frequency converter allows to change the speed of the paddle wheel (from 0 to 12 rpm), which affects the difference between the level in the suction section h_2 (m) and discharge section h_1 (m) of the open tank, and consequently the flow of the culture medium through the transparent tubes.

Each HHT PBR was equipped with probes to continuously monitor dissolved oxygen (DO), pH, and temperature. Each HHT PBR had 11.7 m³ of culture medium. HHT PBR were operated in a semi-batch mode, where every day at 5.00 am 2.3 m³ of culture medium were discharged from the HHT PBR to a settler. Subsequently, at 7.00 am 2.3 m³ of a mixture of municipal and agricultural wastewater were fed to each PBR from a homogenization tank. A detailed description of the HHT PBR function can be found in Díez-Montero et al. (2020).

2.2. Experimental campaign for performance evaluation of the photobioreactor

Two intensive sampling campaigns were carried out in late winter (19th - 22nd March 2018) and in mid spring (8th - 11th May 2018). During both sampling campaigns, 8 samples were collected from the tested HHT PBR every day at three hours intervals, in order to monitor the daily and hourly variations of the main parameters in the HHT PBR. Samples of the culture medium were collected from one open tank, which was considered representative of the bulk volume of the PBR, stored at 4 °C and transported to the laboratory for determination of Total Suspended Solids (TSS) and nutrients: orthophosphate, ammonium, nitrate and nitrite. In addition, grab samples of the influent wastewater were collected from the homogenization tank from which the PBR was fed, stored at 4 °C and transported to the laboratory for determination of the same parameters. The measured data are described more in detail in Díez-Montero et al. (2020).

The HHT PBR operation was monitored by means of online sensors of temperature (Cambell Scientific Inc., USA), DO (Neurtek, Spain) and pH (Hatch Lange SL., Spain) installed in one of the two open tanks. TSS

concentration was analyzed following Standard Methods for the Examination of Water and Wastewater (American Public Health Association, 2005) and ammonium nitrogen was analyzed according to the methods described in Solórzano (1969). Phosphate, nitrate nitrogen, and nitrite nitrogen were measured by means of an ion chromatograph DIONEX ICS1000 (Thermo Fisher Scientific Inc., USA). The intensity of solar radiation was gathered from records of the Catalan Meteorological Service (Servei Meteorològic de Catalunya, 2018).

2.3. Numerical model setup

A CFD model of the horizontal tubes was created to simulate the movement of microalgal cells inside the tubes. Subsequently, the simulated cell movement was integrated into the BIO ALGAE biokinetic model. The multi-physical model resulting from such integration is able to consider the influence of the hydrodynamic conditions of the culture medium on the production of microalgae biomass, so it was simulated under different operating conditions (variation of the mean flow velocity of the culture medium in the tube from 0.19 to 0.37 m s⁻¹).

2.3.1. CFD hydrodynamic model

The geometry of the PBR tubes was created in the ANSYS FLUENT CFD 19.1 software. The fluid flow was simulated as turbulent flow using the Large Eddy Simulation (LES) turbulence model. Momentum, mass, energy, and other parameters are transmitted mainly by large eddies. The LES method is based on the solution of large eddies as spatially and time-dependent formations, that can be captured and filtrated. Turbulent eddies of small scales, which are generally isotropic, are modeled by subgrid models and removed by filtering the turbulent field. Compared to Reynolds Averaged Navier-Stokes (RANS) models, the mesh of the LES must be significantly finer, and the calculations are much more computationally intensive. Mesh needs to be very fine, especially in areas next to the walls in all three directions since the large scale eddies in the turbulent spectrum are geometrically very small there. The control equations used in the LES models are obtained by filtering the non-stationary Navier-Stokes equation. The filtering process removes eddies that are smaller in scale than the width of the filter or the resolution of the used mesh. The equations adjusted in this way control the dynamics of the large eddies (ANSYS Inc. US, 2018).

To simplify the calculations, the three-dimensional CFD geometry of the modeled tube was modified to just 1 m length and a fully developed flow was defined for the inlet. The mesh consisted of 1375,358 elements with a 1 mm maximum size and the structure was modified to be significantly finer in the area around the transparent walls of the tube. An inflation function consisting of 5 layers with a growth value of 1.2 was used to simulate the viscous sublayer and the buffer layer in turbulent flow regime. Four mesh densities were used for analysis of mesh-sensitivity (856,731; 1375,358; 1957,903; 2437,581 elements). The selected mesh was the optimal in terms of accuracy and computational time and ensures mesh-independent results as well. Computations were performed until the calculation converged at a residue of 10^{-5} between two iterations. The mesh quality was checked using skewness reaching the value of 0.21 and orthogonal quality of 0.35. To simulate the hydrodynamic conditions in a modified 3D tube model, the inlet was defined by the inlet velocity according to the selected operating parameters in the HHT PBR. Fig. 2 shows the boundary conditions in developed mesh for the proposed PBR tube. The developed CFD model was calibrated and validated based on the experimental measurements of the flow rate of the culture medium in HHT PBR tubes. The operating conditions used for preliminary calibration and validation of the CFD model are shown in Table 1. For the preliminary calibration and validation of the hydrodynamic model, the method of comparing velocity profiles, which was applied in the study by Belohlav et al. (2021).

Selected configurations were used to study the minimum and maximum flow rate of the HHT PBR. The culture medium level difference Δh (m) resulting from the medium level in the open tanks h_1 (m)

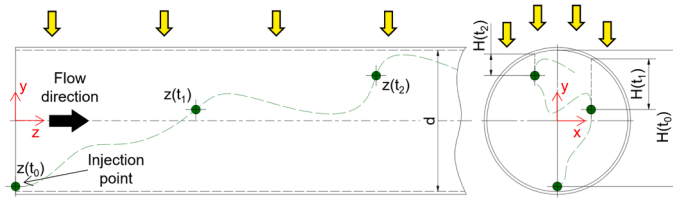


Fig. 3. Scheme of the particle tracking in transparent tube, the yellow arrows represent the direction of the light radiation, $H(t_i)$ indicates the distance of the particle from the irradiated wall at time t_i , $z(t_i)$ represents the position of particle according to axial z -coordinate at time t_i , d is the maximum depth of culture medium in PBR tube.

Table 2

Parameters of integrated multi-physical cultivation model.

Configuration	Re (-)	$\bar{u}$ (m s ⁻¹)	H (m)	d_{tank} (m)
A	23,700	0.19	0.116	0.260
B	31,200	0.25	0.099	0.320
C	46,200	0.37	0.074	0.325

and h_2 (m) was changed by the variation of the paddle wheel speed rotation, resulting in three different configurations (A, B, C). The mean flow velocity $\bar{u}$ (m s⁻¹) increased with the culture medium level difference. Reynolds number Re was calculated as a function of the tube diameter d (m), mean velocity $\bar{u}$ (m s⁻¹), and kinematic viscosity at average culture medium temperature 20 °C ν (m² s⁻¹) in the form

$$Re = \frac{\bar{u} \cdot d}{\nu} \quad (1)$$

The particle tracking function was used to monitor the movement of microalgal cells in the culture medium during its flow in the tube. Moreover, the distance of particle from the irradiated wall H (m) was simulated as well. The original BIO_ALGAE cultivation model (developed by Solimeno, Parker, et al., (2017)) uses the Lambert-Beer's law to define the average light intensity in the culture medium according to the thickness of the culture medium layer. However, this approach does not take into account the effect of mixing the culture medium layer, which ensures the movement of microalgae cells between the dark and light zones formed in the layer. In this study, a local light intensity model was included in the CFD model in order to take into account the hydrodynamics of the system and its influence in the radiation received by the cells. At this aim, the depth of the system has been replaced by the actual distance of the cells from the irradiated surface of the PBR. The principle of the simulation is shown in Fig. 3.

The selected particle parameters were adjusted according to averaged properties of microalgal cells (Belohlav and Jirout, 2019). Specifically, the density of microalgae cells was 1010 kg m⁻³ and the cell diameter is 5 μ m (microalgae cells are represented by a sphere geometry). The injection point was chosen according to the possibility to simulate the movement of a particles that occur in the area of the tube that is less irradiated by the light source (most unfavorable conditions). Since the tube is irradiated only from the upper direction, the place directly below the axis of the tube is furthest from the irradiated surface and suffers the most significant reduction of incident light radiation. The injection point was chosen at a distance of 0.5 mm from the bottom wall of the tube so the movement of the cell at the beginning of the flow was not affected by the laminar or transition area, which is formed near the wall of the tube (Belohlav et al., 2020). The single injection method was chosen for the entry of particles into the tube and the injection point was located at a distance of 0.062 m from the tube axis, i.e. close to the bottom of the tube. The coordinates of the particle injection point are shown in Table 2.

2.3.2. Biokinetic model

The biokinetic model BIO_ALGAE is an integrated mechanistic model that combines the most important physical, chemical, and biokinetic processes of microalgae and bacteria in microalgae-based wastewater treatment systems. The model was developed at the Universitat Politècnica de Catalunya by the GEMMA research group (Solimeno et al., 2017b). The BIO_ALGAE model consists of a combination and adaptation of the River Water Quality Model 1 (RWQM1) and a modified Activated Sludge Model No. 3 (ASM3). The BIO_ALGAE model was implemented in COMSOL Multiphysics™ version 5.4. The HHT PBR geometry was represented in a 1D domain divided into two parts: the open part represents the open retention tanks, and the closed part represents the tubes. The size of both parts corresponds to the entire volume of culture medium.

The model includes the most important operating factors and parameters such as temperature, photorespiration, pH, light radiation, light attenuation, and transfer of gases to the atmosphere. The model has 6 particulate and 13 dissolved components, which are implicated as variables in the physical, chemical, and biokinetic processes. A comprehensive description of the model, including the model equations and the matrix of stoichiometric parameters, can be found in Solimeno, Acien, et al., (2017). The growth of microalgae depends on the intensity of light radiation, temperature, which are the most limiting factors in cultivation systems. However, the model can describe the dependence of microalgae growth on the utilization of nutrients as well. The model needs to be carefully calibrated based on the maximum growth rate of microalgae μ_{ALG} , the maximum growth rate of heterotrophic bacteria μ_H , the decay of heterotrophic bacteria $k_{death,H}$, and the transfer of gases to the atmosphere K_{a,O_2} , K_{a,CO_2} , and K_{a,NH_3} (Solimeno et al., 2017a). Photorespiration is also implemented in the model, which can describe the accumulation of dissolved oxygen in closed culture systems. According to this, it is possible to monitor in which part of the PBR the produced microalgae cells may be inhibited due to the excessive amount of produced oxygen (Solimeno and García, 2019).

2.3.3. Model of light attenuation

The intensity of incident light to PBR decreases exponentially as it passes through the culture medium. In the BIO_ALGAE model, the average intensity of light radiation I_{av} (W m⁻²) acting on microalgal cells is defined by Lambert-Beer's law, which considers the effect of operational parameters (incident light intensity, microalgae and bacteria biomass concentration, or depth of culture medium in PBR) (Huesemann et al., 2016).

This model does not consider the intensity of culture medium mixing as it flows through the transparent tubes. In cultivation systems that have a wide layer of culture medium (e.g. raceway ponds, column PBRs), the intensity of mixing can affect the efficiency of the whole process of microalgae cultivation. In systems with a thick layer of culture medium and low mixing intensity, only cells that are close to the transparent surfaces of the photobioreactor are sufficiently irradiated. The irradiation of the remaining cells is limited here due to the reduction of the incident light radiation when penetrating the culture medium layer. However, when mixing is intensified, microalgae cells can move from dark zone into the culture medium area, where the intensity of penetrating light radiation is sufficient (light zone). Microalgae cells thus receive the amount of light radiation needed for their growth, which increases overall productivity (Belohlav et al., 2018).

The intensity of incident light radiation on microalgal cells at a given depth I_p (W m⁻²) is influenced by the attenuation of the layer of culture medium above the cell, and the mutual attenuation of microalgal cells (Grima et al., 1997). The local light intensity at a given depth H (m) of the microalgal cell can be expressed by Eq. (2):

$$I_p = I_0 \cdot e^{-K_I \cdot X_C \cdot H} \quad (2)$$

where I_0 (W m⁻²) is the incident light intensity, K_I (m² g⁻¹) is the

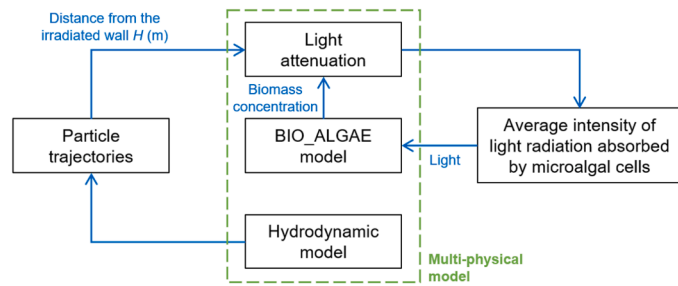


Fig. 4. Coupling multi-physical model methodology.

extinction coefficient, X_C (g m^{-3}) is the sum of particulate components (microalgae biomass and bacteria). Therefore, in the present work, the average intensity of light radiation in BIO_ALGAE model representing general conditions in tube (Lambert-Beer's law) was replaced by the local light intensity at a given depth (Eq. 2), which can consider the hydrodynamic conditions in the cultivation system. The local light intensity I_p can be defined by CFD particle tracking model. If the culture medium is intensively mixed, the microalgae cells in the dark zones are alternately transferred by the vortices of the flowing medium closer to the irradiated area, where the intensity of light is sufficient for microalgae growth. The reflection of the incident light radiation and the mutual shielding are also important parameters for the irradiation of transparent surfaces (Slegers et al., 2013b, 2013a, 2011). In the developed model, the influence of light reflection and mutual shading during the cultivation day is neglected and the dominant influence of the attenuation of the incident light radiation during the penetration of the culture medium is assumed.

2.3.4. Modelling, preliminary calibration and simulation methodology

The developed hydrodynamic model allows to predict the movement of microalgae cells depending on changing hydrodynamic conditions. Changing the trajectory of the flowing cells allows them to reach the irradiated zone of the culture medium layer. Based on the intensity of the incident light and the trajectory of the cell movement, the intensity of light radiation on microalgae cells flowing through the transparent tube of the HHT PBR under selected hydrodynamic conditions can be determined. The intensity of received light radiation subsequently influences the overall production of microalgal biomass. Changing the biomass concentration in the processed microalgae suspension affects the permeability of the culture medium layer to the incident light radiation. The modeling methodology is shown in Fig. 4.

The developed multi-physical model was calibrated using experimental data collected during the winter intensive measurement campaign. The functionality and robustness of the developed model was validated using experimental data from the spring intensive

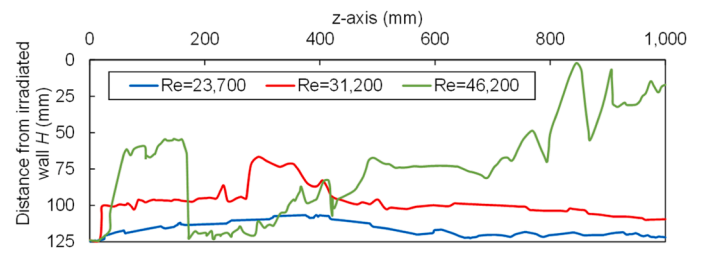


Fig. 6. The average distance of 200 particles from the irradiated wall along the HHT PBR tube, H indicates the distance of particles from the irradiated wall (0 mm means the position of irradiated tube wall, while 125 mm means the maximum distance from irradiated wall) and z -axis indicates the axial dimension of the tube.

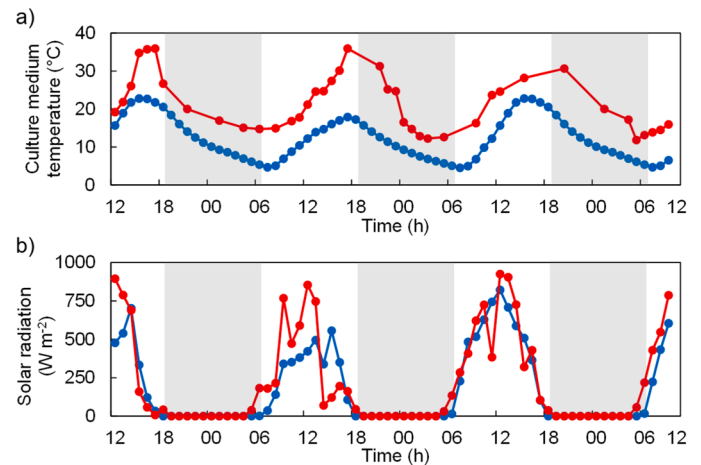


Fig. 7. Measured parameters – a) culture medium temperature and b) solar radiation. Blue represents the winter campaign (March), red indicates the spring campaign (May). The vertical white and grey stripes represent the daylight and night periods.

Table 3
Calibrated parameters.

Parameter	Description	Value (d^{-1})
μ_{ALG}	Maximum specific growth rate of microalgae	1.7
μ_H	Maximum specific growth rate of heterotrophic bacteria	1.3
$k_{death,H}$	Inactivation constant of heterotrophic bacteria	0.3
K_{a,O_2}	Mass transfer coefficient for oxygen	3.7
K_{a,CO_2}	Mass transfer coefficient for carbon dioxide	3.0
K_{a,NH_3}	Mass transfer coefficient for ammonia	0.7

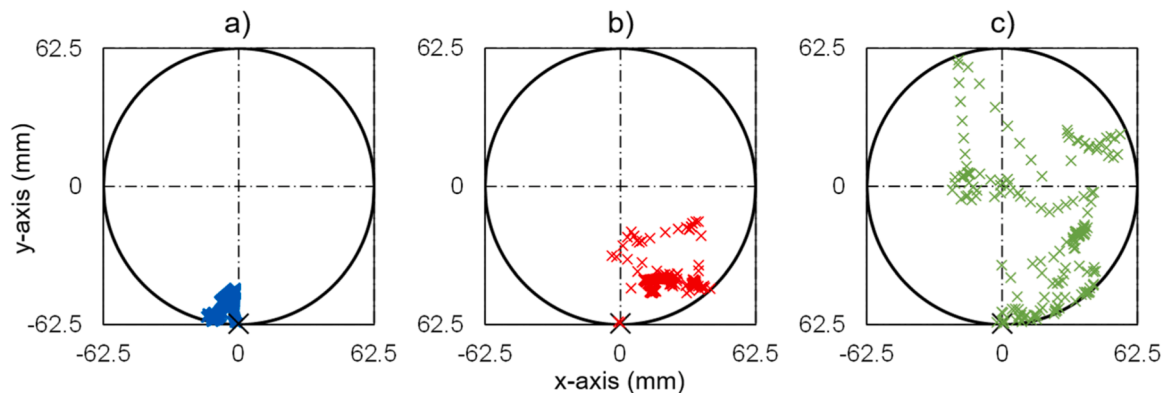


Fig. 5. Particle tracking in HHT PBR tube. Cross-sectional view of the tubes (black circles) – a) $Re = 23,700$, b) $Re = 31,200$, c) $Re = 46,200$. Black cross indicates particle injection point. The coloured crosses represent the position of the particles at a distance of 1 m from the injection point.

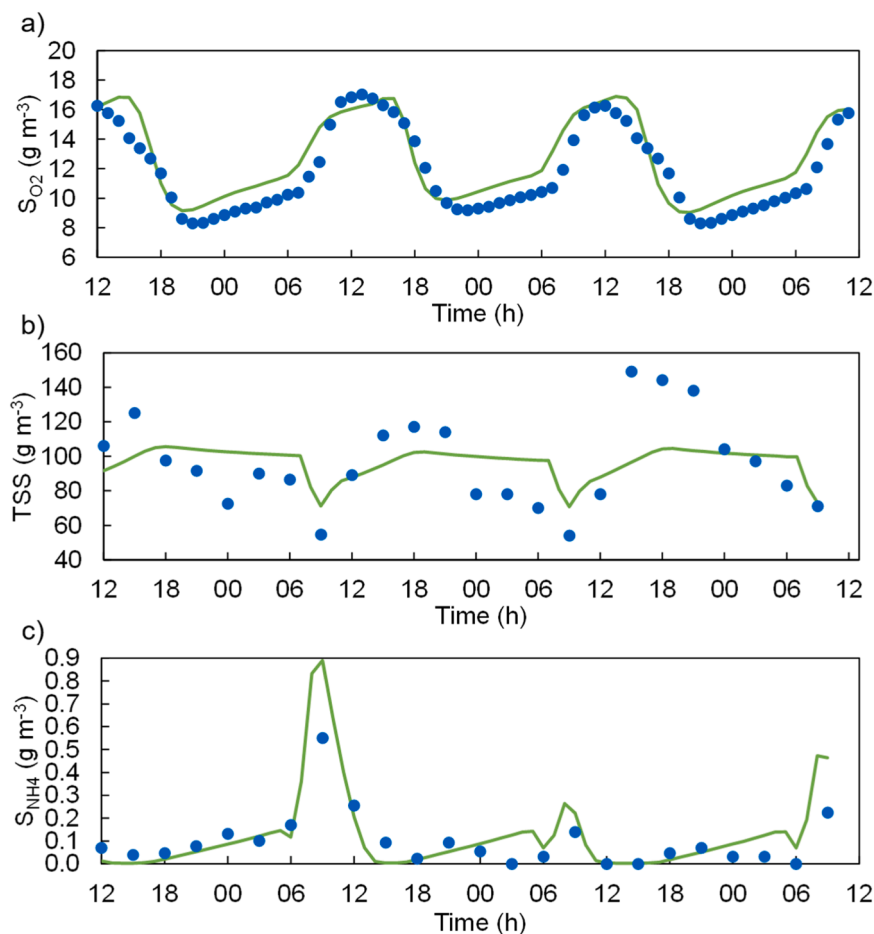


Fig. 8. Calibration of the model of HHT PBR – a) dissolved oxygen, b) total suspended solids and c) ammonium nitrogen. Simulated (green line) and experimental (blue dots) results obtained during the winter campaign.

measurement campaign. The validated multi-physical model allows to simulate the effect of changing hydrodynamic conditions on the intensity of incident light radiation on microalgal cells, and influence on microalgal biomass production in HHT PBR.

3. Results and discussion

3.1. Particle tracking

Using the developed validated CFD model (Belohlav et al., 2021), the movement of particles was monitored under the three selected operating conditions of HHT PBR (Table 1). The movement of the particles (200 particles were simulated for each operating condition settings) during their flow through the tube shows the dispersion of motion strictly related with the increasing flow velocity in the tube (Fig. 5).

The distance of particle from the irradiated surface of the tube H (m) can be derived from particle tracking distribution. In HHT PBR, depending on the incident radiation and the concentration of microalgae in the culture medium, only the upper part of the tube is irradiated. If the culture medium is insufficiently mixed, the particles moving in the lower part of the tube will not be irradiated as the incident radiation is scattered by the layer of culture medium and by the other cells in the culture medium. The derived average distance of the 200 simulated particles from the irradiated wall H (m) during the flow through the tube is shown in Fig. 6. From the comparison, it can be seen that at lower flow velocities (blue line), the particles are still moving close to the initial position. As the flow rate increases (red and green lines), the particles more often get closer to the irradiated area of the tube. Intensification of

mixing allows the transition of particles from the dark zone to the light zone of the tube. At the same time, it can be assumed that particles in the light zone can also repeatedly move into the dark zone and back again, which can have a positive effect on biomass production (Abu-Ghosh et al., 2016).

3.2. Preliminary calibration and validation of multi-physical model

The multi-physical model integrating the hydrodynamic and biokinetic model was calibrated using the measured data of DO, TSS and ammonium nitrogen from the winter sampling campaign (Díez-Montero et al., 2020). The outputs of the biokinetic model are particularly sensitive to the maximum specific growth rate of microalgae, and mass transfer coefficients for oxygen and carbon dioxide. Solimeno et al. (2016) in his study described in detail the parameter sensitivity analysis. The velocity of culture medium in transparent tubes was set at 0.25 m s^{-1} ($Re = 31,200$) during the measuring campaigns, which corresponds to configuration B (Table 1). Considering the complexity of the multi-physical model calculation, the average value of the distance of simulated 200 particles from the irradiated wall H (m) was used in the model. The model also includes the average depth of the culture medium in the open tanks d_{tank} (m). The average value of d_{tank} is specified based on the level difference on both sides of the tank, i.e., the average of h_1 and h_2 shown in Table 1. The values of H and d_{tank} for the various operating configurations are shown in Table 2.

The influent wastewater was characterized by average TSS concentration of 72 mgTSS L^{-1} in winter. Regarding the evolution of ammonium nitrogen, the concentration ranged between 1.5 and 3.5 mgN L^{-1} .

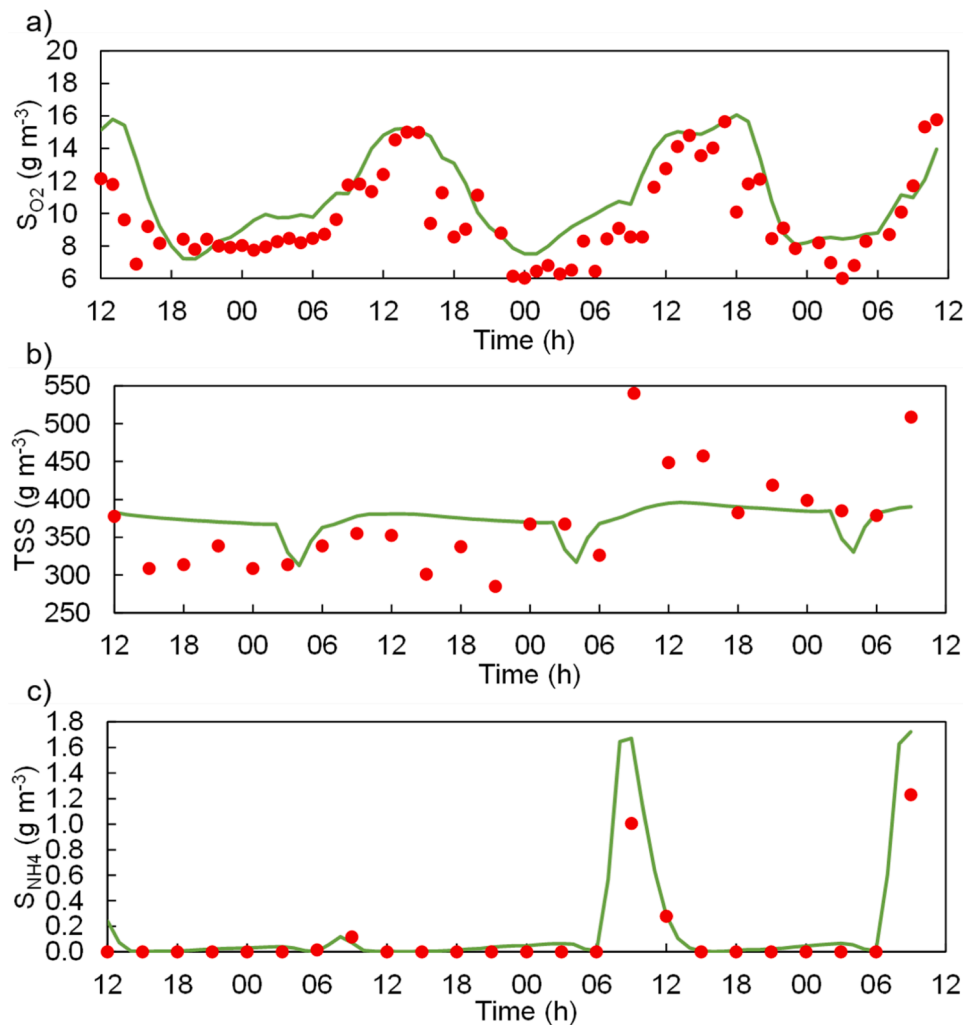


Fig. 9. Validation of the model of HHT PBR – a) dissolved oxygen, b) total suspended solids and c) ammonium nitrogen. Simulated (green line) and experimental (red dots) results obtained during the spring campaign.

The measured values of the culture medium temperature in HHT PBR (Fig. 7a) oscillated between 5 and 20 °C in winter and 15–35 °C in spring, while the intensity of solar radiation (Fig. 7b) reached 900 and 800 W m⁻² at around 12:00 p.m. in spring and winter, respectively. These records of the solar radiation and medium temperature were used as input data in the multi-physical model as well. After running the simulation, it is necessary to wait for the microalgae cultivation process to stabilize in the biokinetic model. The cultivation campaign was set to a repetitive length of tens of days in the simulation environment to allow the culture system to stabilize. After all operating parameters were stabilized, data from the last three days were used to calibrate the model according to the measured experimental data.

The numerical model was calibrated by comparing the simulated dissolved oxygen, total suspended solids, and ammonium nitrogen content with the experimental data obtained in the winter campaign. The parameters μ_{ALG} , μ_H , $k_{death,H}$, K_{a,O_2} , K_{a,CO_2} , and K_{a,NH_3} were adjusted in order to calibrate the model. The resulting values of these parameters are shown in Table 3. The following parameters were used for calibration: dissolved oxygen S_{O_2} (gO₂ m⁻³), total suspended solids TSS (gCOD m⁻³), and ammonium nitrogen S_{NH_4} (gNH₄⁺-N m⁻³). The resulting curves of the calibrated model are shown in Fig. 8.

Calibration was performed to minimize the root mean square error (RMSE). The RMSE of the calibrated model for each component was RMSE_{S_{O₂}} = 1.36 gO₂ m⁻³, RMSE_{TSS} = 12.61 gTSS m⁻³ and RMSE_{S_{NH₄}} = 0.06 gNH₄-N m⁻³. The higher value of RMSE_{TSS} is due to the model

considering respiration at the end of the night, resulting in a steep drop in biomass. The decrease in experimentally measured TSS during the night cycle is due to low oxygen production in the culture medium, which does not help to keep the biomass buoyant.

The model was validated according to the experimental results from the spring campaign. The influent wastewater characteristics (115 mgTSS L⁻¹), initial conditions, culture medium temperature (Fig. 7a), and solar radiation (Fig. 7b) were slightly different compared to those of the winter campaign. The simulations using the calibrated model, with no further modification of the model parameters, matched quite well the experimental data, as shown in Fig. 9. The global error of the simulations was slightly higher in comparison with calibration. Validation of dissolved oxygen is shown in Fig. 9a with RMSE_{S_{O₂}} = 2.40 gO₂ m⁻³. Validation of TSS and ammonium nitrogen are shown in Fig. 9b and Fig. 9c. The RMSE value of each component was RMSE_{TSS} = 34.00 gTSS m⁻³ and RMSE_{S_{NH₄}} = 0.10 gNH₄-N m⁻³. TSS values were influenced by influent parameters in addition to respiration and oxygen production during the night. The model is not sensitive enough to capture such a step change and showed only a gradual increase.

3.3. Influence of mixing on light intensity and biomass production

The flow velocity and the associated mixing of the culture medium in the transparent tubes affected the average intensity of light radiation I_{av} acting on microalgal cells. A comparison of the I_{av} for the different

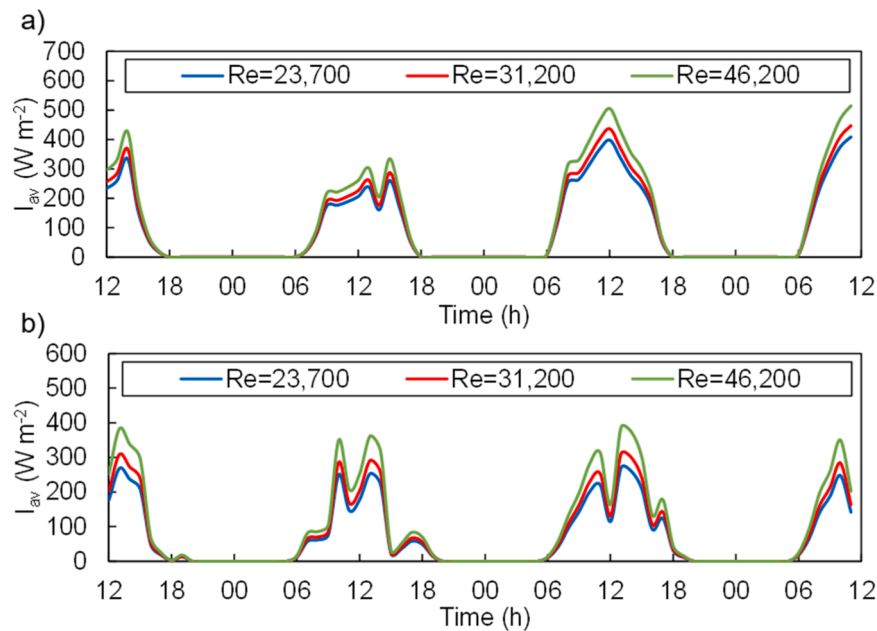


Fig. 10. Average intensity of light radiation corresponding to different Reynolds numbers – a) simulation of winter period cultivation and b) simulation of spring period cultivation.

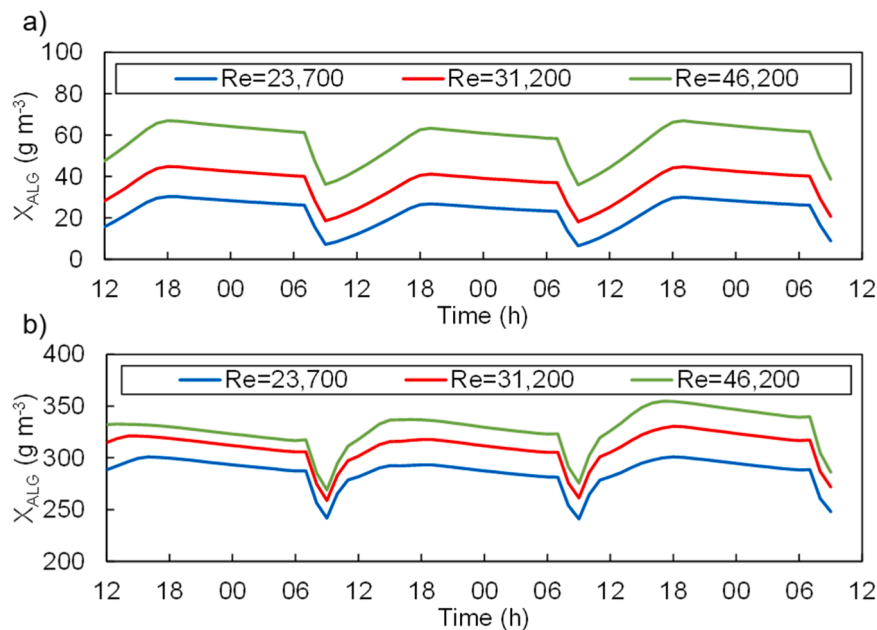


Fig. 11. Microalgae production – a) simulation of winter period cultivation and b) simulation of spring period cultivation.

operating conditions in both seasons, winter and spring, is shown in Fig. 10.

The increase of I_{av} due to the increasing flow rate of the culture medium in the transparent tube is evident. A comparison of the I_{av} obtained by the simulations in the winter and spring seasons shows that in spring the I_{av} values are lower than in the winter campaign, although the incident light intensity I_o was higher during the spring campaign. This is due to the higher concentration of microalgae and bacteria in the culture medium (TSS ranged between 54 and 149 g m^{-3} in winter and between 285 and 540 g m^{-3} in spring, as can be observed in Figs. 8b and 9b, respectively), which reduces the light penetration into the culture medium and eventually the average light intensity received by the cells.

Regarding the influence of the mixing conditions and flow rate of the

culture medium inside the tube, the results of the simulations showed that the concentration of microalgae increased according to intensified mixing conditions, as it can be seen in Fig. 11. When the flow regime was increased from $Re = 23,700$ – $31,200$ and $46,200$ in winter season, the average concentration of microalgae in the culture medium increased by 60% and 151% (14 and 35 g m^{-3}), respectively. While the same change in flow regime in the spring period resulted in increases of 8% and 14% only (24 and 41 g m^{-3}). The lower concentration increase during the spring period may be caused by the limited amount of nutrients in the processed culture medium (Díez-Montero et al., 2020).

Note that in this study the slight increase in the concentration of microalgae is influenced by the overall low production of microalgae in the system due to nutrients limitation. The concentration of microalgae

in the culture medium reached 67 g m^{-3} during the winter season and 355 g m^{-3} during the spring season. Under the conditions of this study, the culture medium had a low concentration of microalgae and the light was not a clear limiting factor. Nevertheless, this result demonstrates and quantifies the influence of hydrodynamics on biomass production, and the integrated multi-physical model could be applied for simulating other tubular photobioreactors or influent wastewater characteristics.

Díez-Montero et al. (2020) described that the concentration of microalgae in the HHT PBR was limited by a lack of nutrients. More precisely, it was a lack of phosphorus, which was deficient throughout the whole year in the influent wastewater, thus limiting the biomass production. Similar cultivation systems fed with higher loads of nutrients usually reach microalgae concentrations over 1000 g m^{-3} (Olivieri et al., 2014). The mixing intensity of the culture medium would have a more intensive effect in systems working with the significantly greater thickness or depth of the culture medium, such as raceway pond or column PBRs. These systems usually work with a wide layer (0.3 m), which can be difficult to irradiate even with low concentrations of microalgae in the culture medium.

It is also necessary to consider the operating costs and the costs associated with maintenance during the selection of operating conditions in HHT PBR. As the flow rate of the culture medium in the tubes increases, the consumption of electricity increases as well, which is associated with the rotational speed of the paddle wheel. As the speed of rotation increases, also consequential wear of the engine can occur, which leads to increased maintenance costs. Conversely, in the case of the increasing flow rate of the culture medium in the tubes, the effect of increasing shear stresses on the wall can lead to biofilm removal, which has a strong effect on the culture medium irradiation and microalgae production in closed cultivation systems (Belohlav et al., 2020).

4. Conclusions

A multi-physical model coupling a mechanistic biokinetic BIO-ALGAE model and a computational fluid dynamics (CFD) hydrodynamic model was developed. The applicability of the developed multi-physical model was calibrated and validated using experimental data from two intensive measuring campaigns. Using the developed model, it was possible to investigate in detail the effect of hydrodynamic conditions on the ability to illuminate the culture medium layer and to supply sufficient light radiation to the microalgae cells. Subsequently, it was possible to predict the effect of changing the operating conditions of the hybrid horizontal tubular photobioreactor (HHT PBR) on the microalgae yield. The results showed that increasing flow rate provided more intense mixing of the culture medium, which resulted in the microalgae cells being more frequently exposed to incident light. With the increased flow rate of the culture medium, microalgal biomass production increased up to 151% compared to the initial operating conditions. Due to the limitations of the HHT PBR design (engine power, paddle wheel loading), higher production rates could not be used for calibration and validation in the present study. Microalgae concentrations in the experimental system were low ($67\text{--}355 \text{ g m}^{-3}$), which was attributed to nutrients limitation in the influent wastewater. Due to the low concentration of microalgal cells in the culture medium, no significant reduction of incident light radiation occurred when penetrating the culture medium layer. Thus, the culture medium layer in the transparent tubes could be illuminated by light of at least the minimum intensity required for the growth of microalgal cells. For this reason, there was no significant increase in microalgal biomass production when the mixing of the culture medium was intensified. Nevertheless, this study demonstrates and quantifies the influence of hydrodynamics on biomass production, and the model could be used for design of other tubular photobioreactors or for characterization of proper influent wastewater characteristics. Optimization of hydrodynamic conditions would have a significant impact in cultivation systems processing highly concentrated culture media (thin-layer or thin-film flat panel photobioreactor). It can

be assumed that hydrodynamic conditions would also have a significant effect in systems processing large layers of culture medium (raceway pond, flat-panel, column, or airlift photobioreactor) that would be difficult to illuminate with a light source.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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